

How to begin with project

Step1:

Create new project in vivado

Name it 'ahb_spi_bridge_ip_creator'

Click Next till board selection page

Search board Pynq-z2 or search Part no. xc7z020clg400-1

Finish

Step2:

Add source files from AHBLite_SPI_Bridge_Project/Vivado_Design_Files/source

Add all 5 source files (ahb_spi_bridge.v, async_fifo.v, axi4lite_to_ahb_bridge.v, spi_master.v, top_pynq.v)

Step3:

After adding files go to Tools -> Create & Package new ip -> keep "Package your current project" selected -> copy the ip location shown something like (/AppData/Roaming/Xilinx/Vivado/./ip_repo) -> Finish

Step4:

In packaging steps, go to Addressing and Memory -> Change base address to 0x40000000 & range to 0x1000 -> enter

Go to review and package IP -> click Package IP

Step5:

Now create new vivado project keep the current project open in background

Name it 'ahb_spi_bridge_ip_user'

Click Next till board selection page

Search board Pynq-z2 or search Part no. xc7z020clg400-1

Finish

Step6:

In project manager go to settings -> IP -> Repository -> Add -> Paste the copied IP location here -> Click OK

Step7:

Now in IP INTEGRATOR -> Create block design -> Name the design 'System'

In the diagram window -> Add -> Search 'top_pynq_v1_0' -> Select it

Similarly search for 'Zynq Processing System' & Select it

Step8:

Now, **Run Block Automation** then **Run Connection Automation**

Right click on **top_pynq_v1_0** block -> Click **Make external**

Right click anywhere in the Diagram window and **Validate design**

Should show Validation Successful

Step9:

Go to source -> Add constraint file from ->

AHBLite_SPI_Bridge_Project/Vivado_Design_Files/constraint

now under Design Sources -> **system(system.bd)** <- Right click on this and **Generate output products** -> Generate -> after this right click again on system(system.bd)

and **Create HDL wrapper** -> After this **system_wrapper** will be visible under Design Sources

Step10:

Now under Address Editor confirm Base address is 0x40000000 & range is 4K

after this directly generate bitstream

after successful bit generation

Open Hardware Manager -> Open target -> Auto connect -> program device -> click program

[Whichever page is not discussed keep it as it is in default selection]